

6(a)(i)	E = deoxyribose ; F = cytosine ;	2
6(a)(ii)	a ring drawn around 1 base, 1 deoxyribose and 1 phosphate on the same strand and passing through the H bonds ;	1
6(a)(iii)	phosphodiester ;	1
6(b)	All 4 triplets must be correct ; CCA CGA TTA GAT GGU GCU AAU CUA	1
6(c)	<i>any two from:</i> removal of introns / non-coding sequences ; leaving, exons / coding sequences (joined) ; ref. to splicing RNA transcript = 1 mark AVP ; e.g. adding a 5' cap / 3' poly A tail	2

Question	Answer	Marks
6(d)	<i>any five from:</i> 1 two codons at a time are exposed (to the large subunit) ; 2 (idea of) a specific tRNA brings a specific amino acid ; 3 a tRNA anticodon binds to the mRNA codon ; 4 complementary base pairing occurs (by hydrogen bonding) ; 5 a second tRNA brings another amino acid (next to the first amino acid) ; 6 peptide bond formation between the two amino acids ; 7 ribosome moves along the mRNA, one codon at a time / and next codon is 'read' ; 8 the first tRNA leaves the ribosome ; 9 AVP ; e.g. first codon is always AUG first anticodon is always UAC first amino acid is always methionine ribosome moves along mRNA in a 5' to 3' direction role of peptidyl transferase eventually a stop codon is reached (and translation stops)	5